

Spectral Transformation for Layer-wise Global Rank Discovery in Federated LoRA for Vision Transformers

Hariharan Ramesh

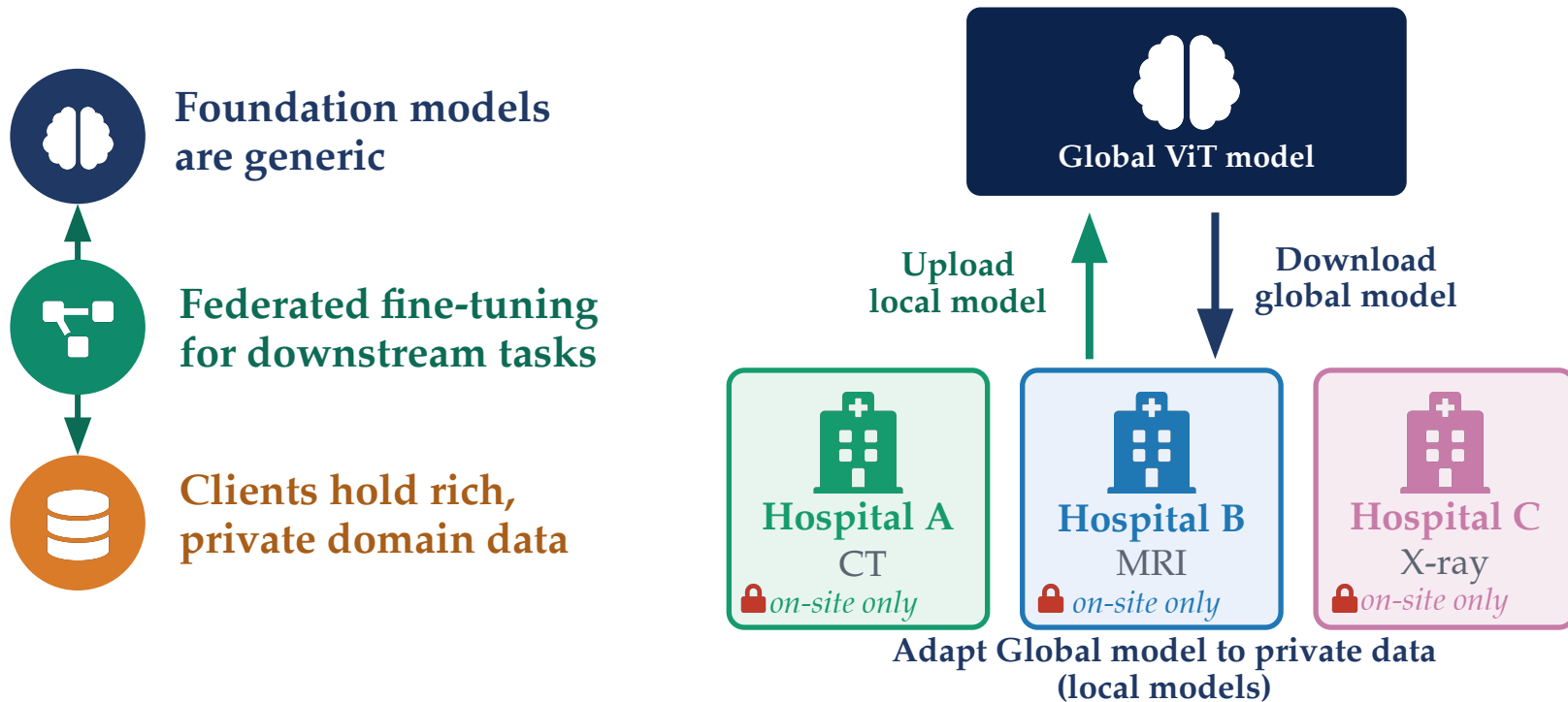
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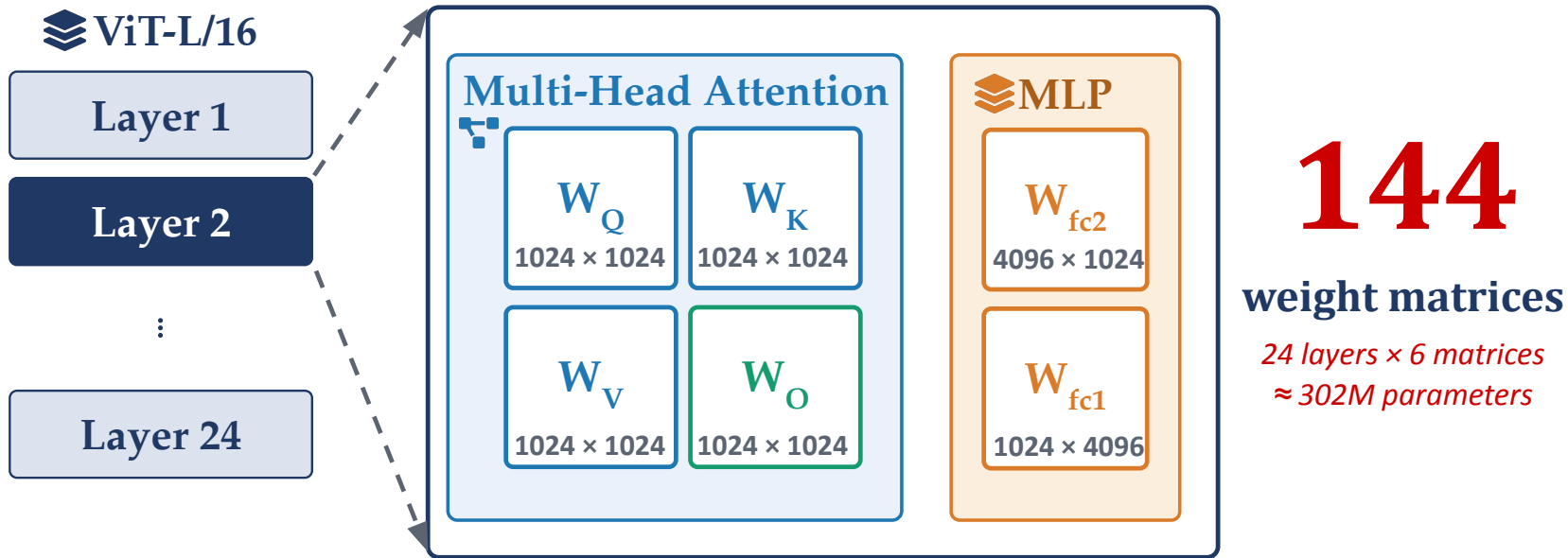
Why Federated Vision Fine-Tuning?

Adapt powerful Vision Transformers to private, domain-specific data, without ever centralizing it



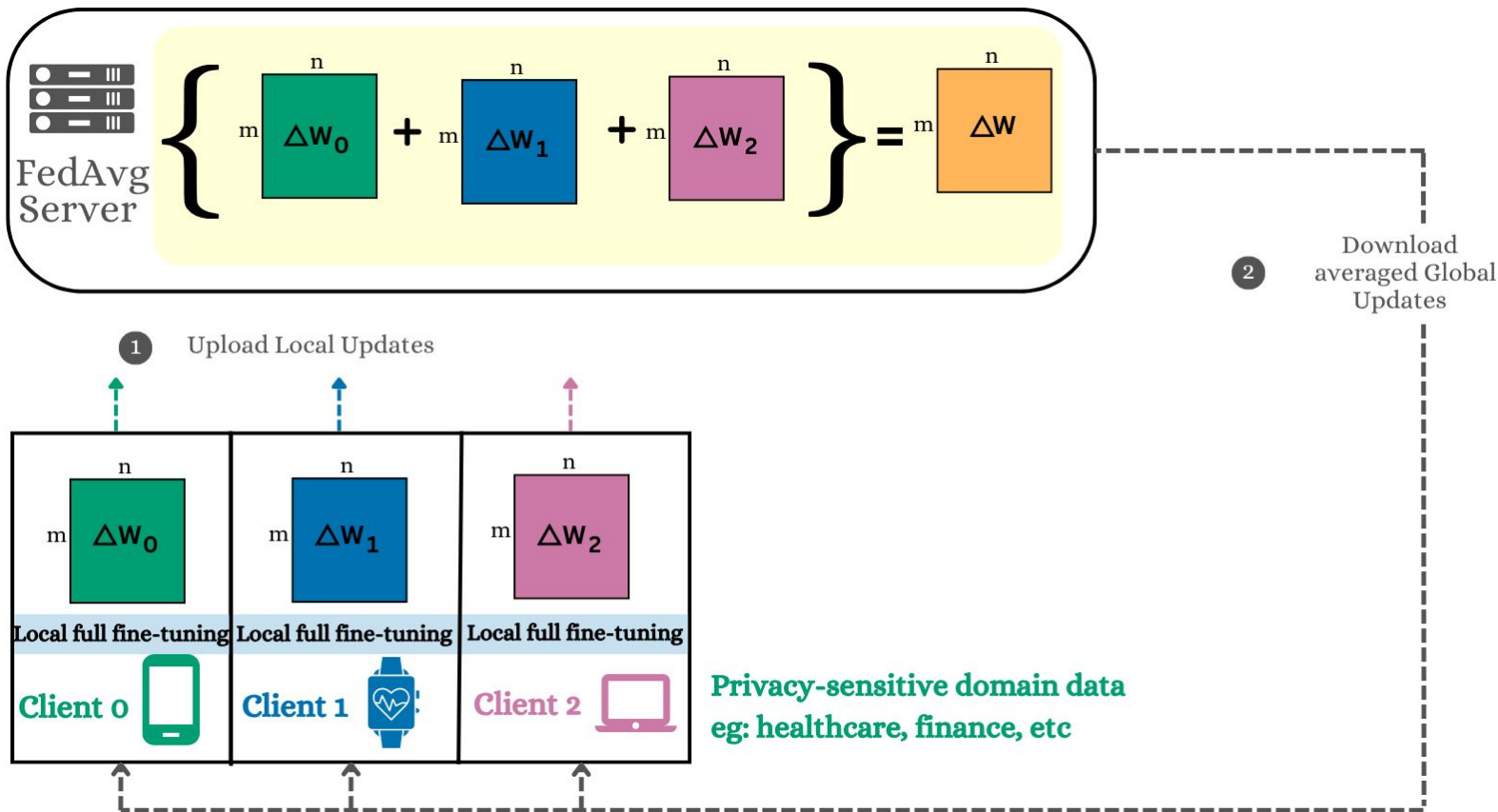
Vision Transformer

Every block of layer holds 6 large weight matrices

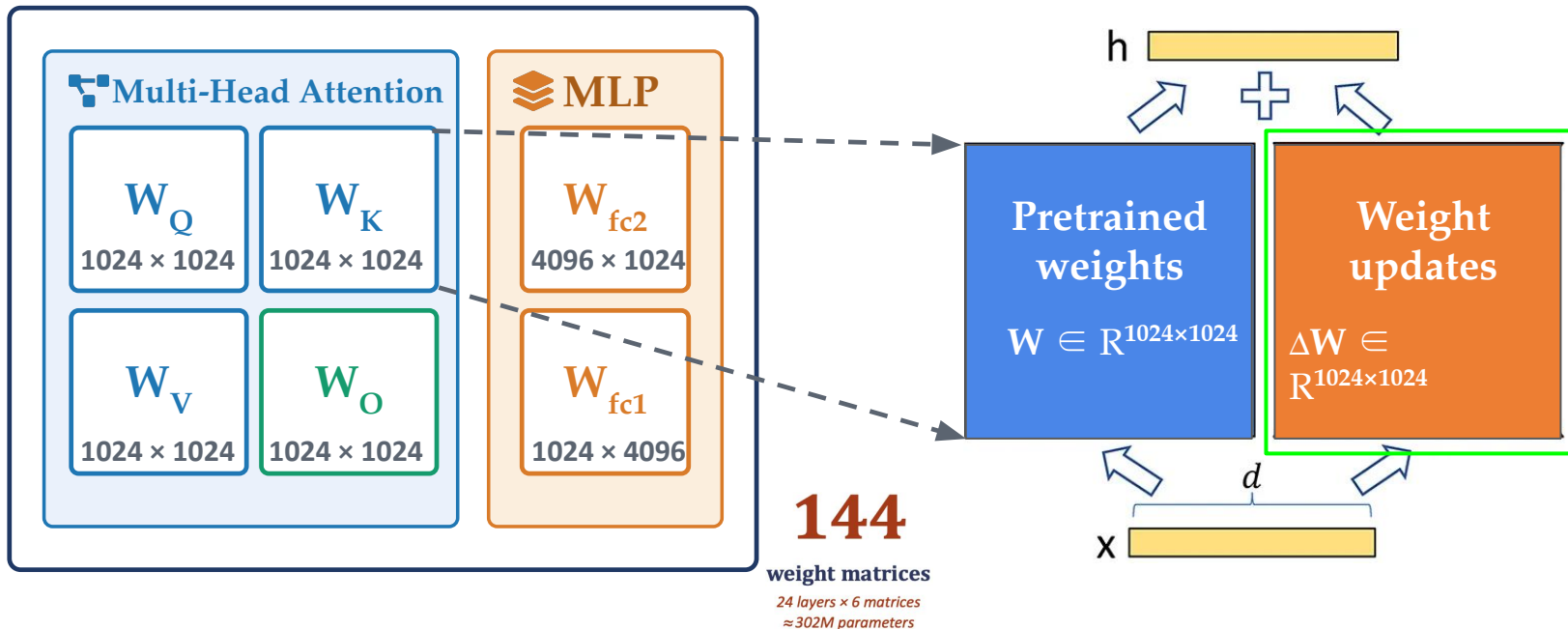


Large, non-uniform weight matrices (1024×1024 in attention, 1024×4096 in the MLP) repeated across all 24 blocks.

Federated Full Fine-Tuning



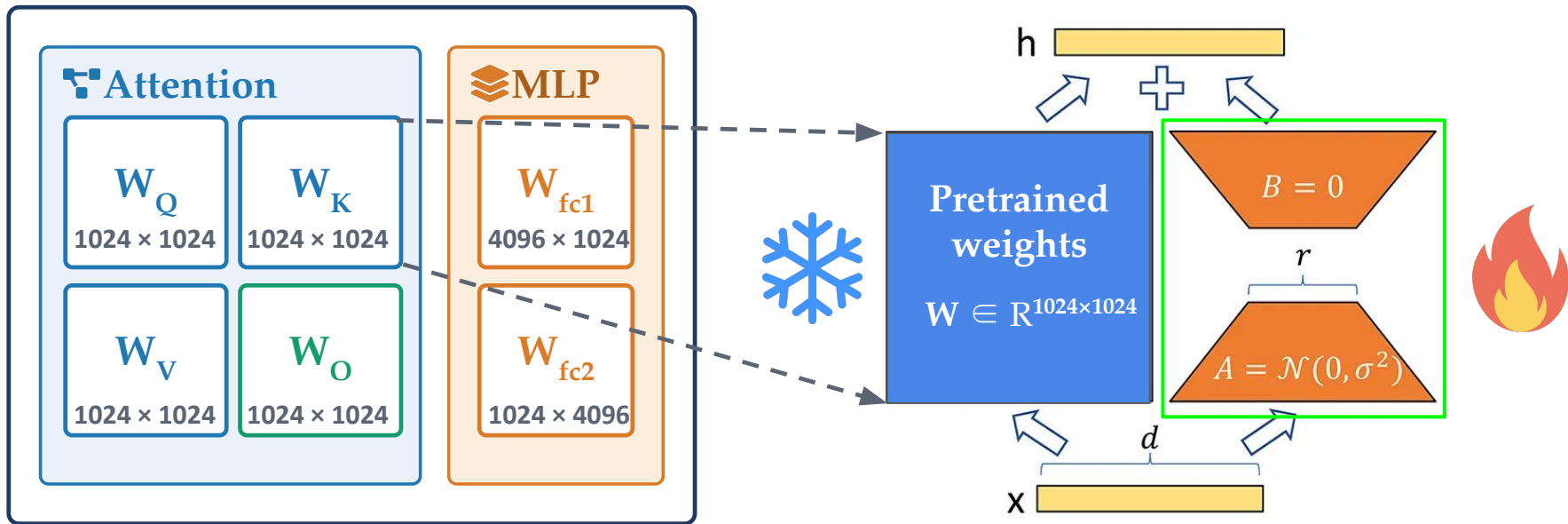
Full-Fine Tuning is Expensive!



Large, non-uniform weight matrices (1024×1024 in attention, 1024×4096 in the MLP) repeated across all 24 blocks.

Low-Rank Adaptation (LoRA)

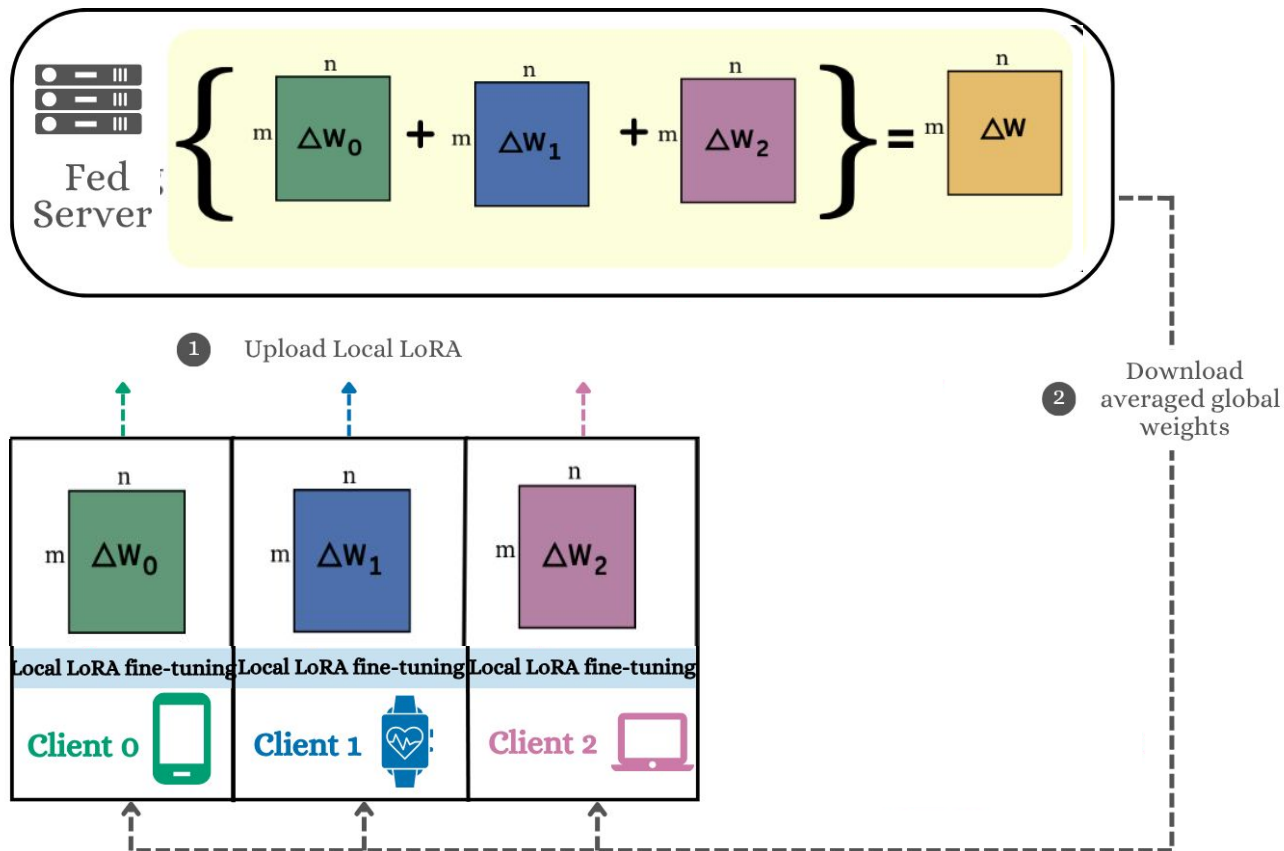
Parameter Efficient Fine-Tuning



LoRA fine-tuning (rank, $r = 8$) creates efficient adapters, $\{B \in \mathbb{R}^{1024 \times 8}, A \in \mathbb{R}^{8 \times 1024}\}$ weight matrices (1024×1024 in attention, 1024×4096 in the MLP) repeated across all 24 blocks

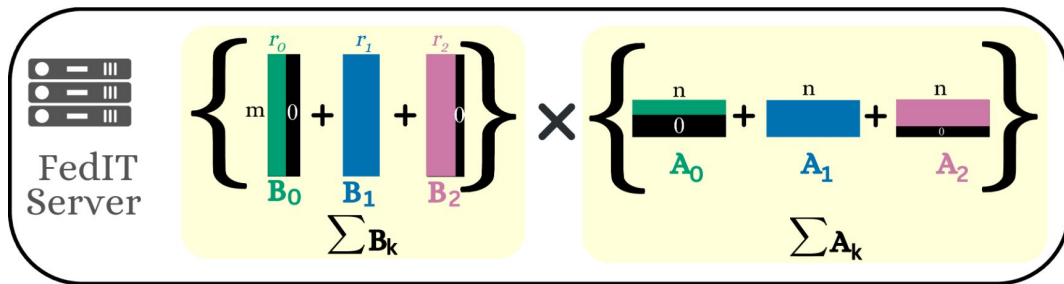
64X reduction in trainable parameters

Federated LoRA Fine-Tuning



Related Work: FedIT (Average)

First Federated Fine-tuning method for Language Models



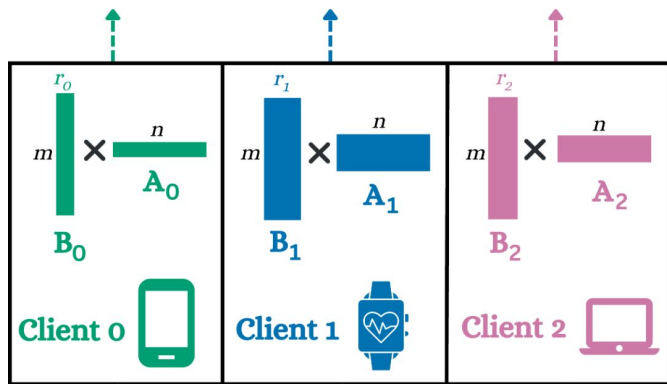
FedAvg applied directly to LoRA adapters

1 Upload Local LoRA

2 Download averaged global LoRA

$$(\sum B_k)(\sum A_k) \neq \sum B_k A_k$$

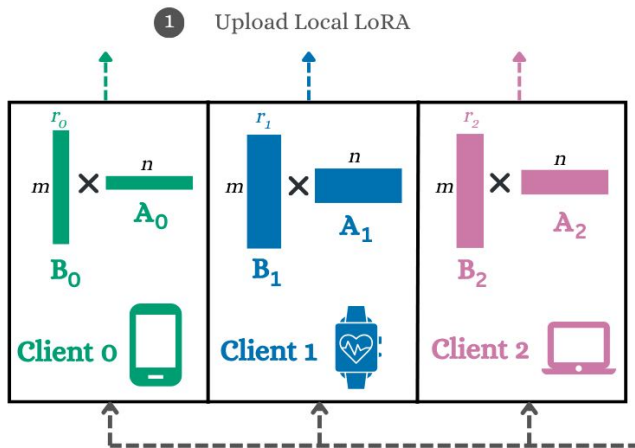
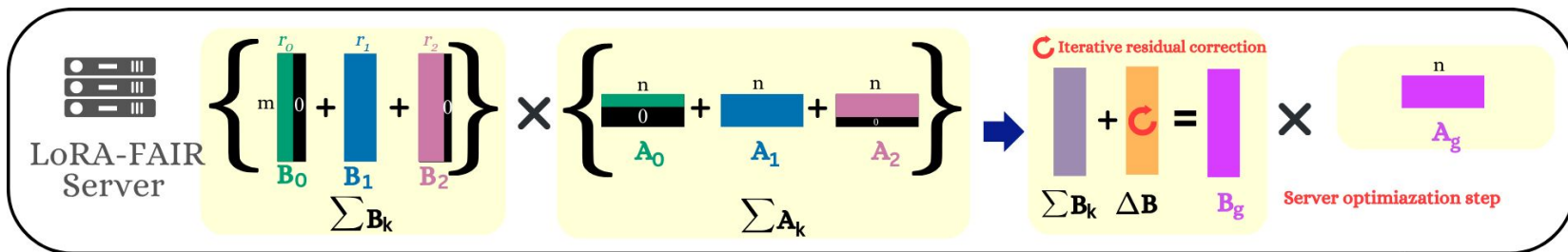
Mathematical inconsistency causing cross-term noise



Cross-Term Aggregation Noise

Related Work: LoRA-FAIR (Average)

First Federated Fine-tuning method for Vision Models



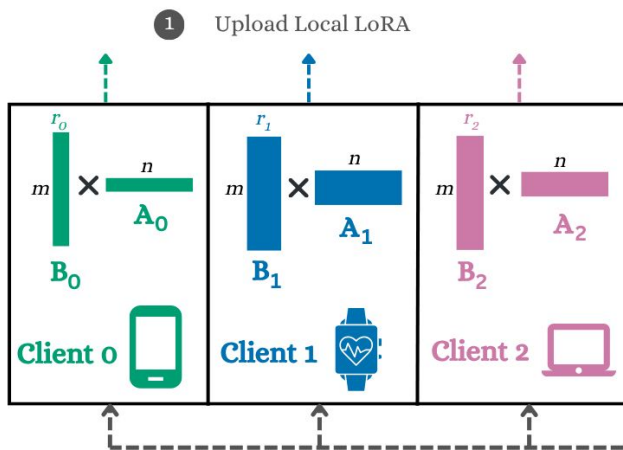
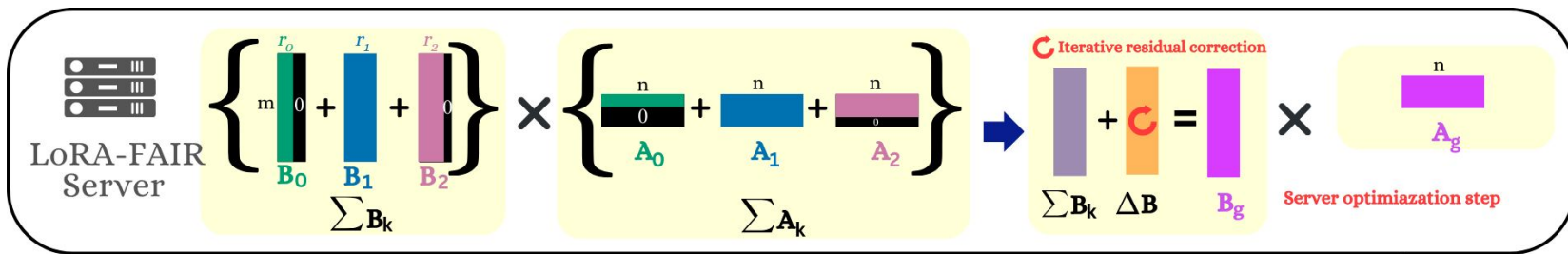
Tries to reduce cross-term noise via
Residual Correction

$$\arg \min_{\Delta B} \underbrace{\mathcal{S}(\Delta W, (\bar{B} + \Delta B)\bar{A})}_{\text{correction}} + \underbrace{\lambda \|\Delta B\|}_{\text{regularization}},$$

Server optimization step is an
ML problem itself!

Related Work: LoRA-FAIR (Average)

High Computational overhead



2 Download averaged global LoRA B_g, A_g

Natively Homogeneous across clients, layers, attention types

Related Work: Average vs Stacking

Averaging methods: FedIT, LoRA-FAIR

$$\left\{ \begin{matrix} r_0 \\ m \end{matrix} \begin{matrix} B_0 \\ 0 \end{matrix} + \begin{matrix} r_1 \\ B_1 \end{matrix} + \begin{matrix} r_2 \\ B_2 \end{matrix} \right\} \times \left\{ \begin{matrix} n \\ A_0 \end{matrix} + \begin{matrix} n \\ A_1 \end{matrix} + \begin{matrix} n \\ A_2 \end{matrix} \right\} = \sum B_k \times \sum A_k$$

✓ Vision support (LoRA-FAIR)

≠

Mathematically accurate aggregation

$$\left\{ \begin{matrix} n \\ m \end{matrix} \Delta W_0 + \begin{matrix} n \\ m \end{matrix} \Delta W_1 + \begin{matrix} n \\ m \end{matrix} \Delta W_2 \right\} = \begin{matrix} n \\ m \end{matrix} \Delta W$$

||

Stacking methods: FLoRA, FLoRIST

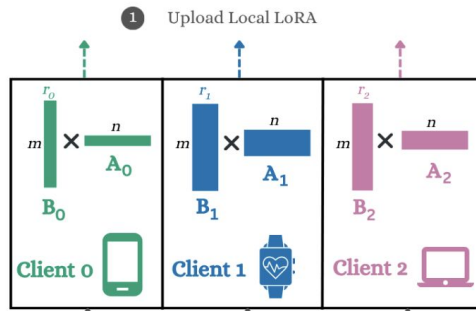
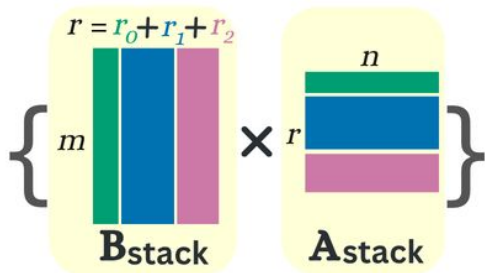
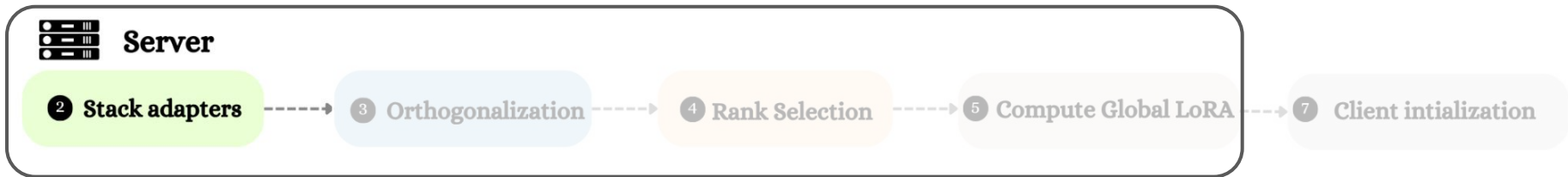
$$\left\{ \begin{matrix} r = r_0 + r_1 + r_2 \\ m \end{matrix} B_{stack} \right\} \times \left\{ \begin{matrix} n \\ r \end{matrix} A_{stack} \right\}$$

🚫 Only applied on LLMs

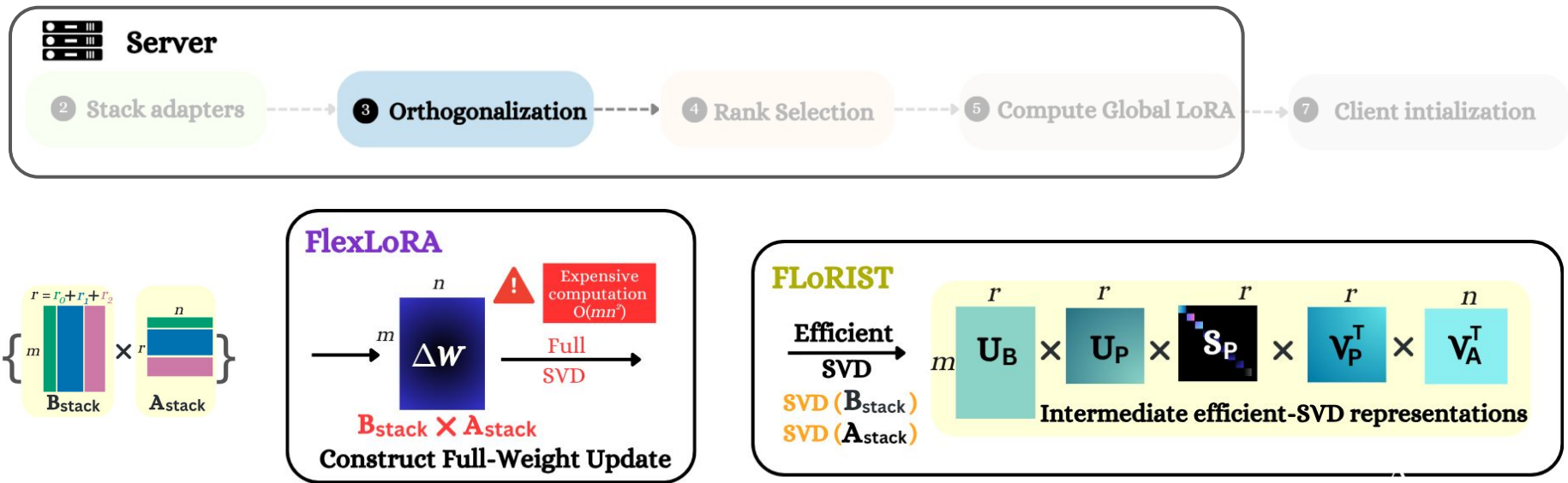


Challenge 1: Stacking based methods haven't been explored in Vision

Related Work: Stacking



Related Work: Stacking

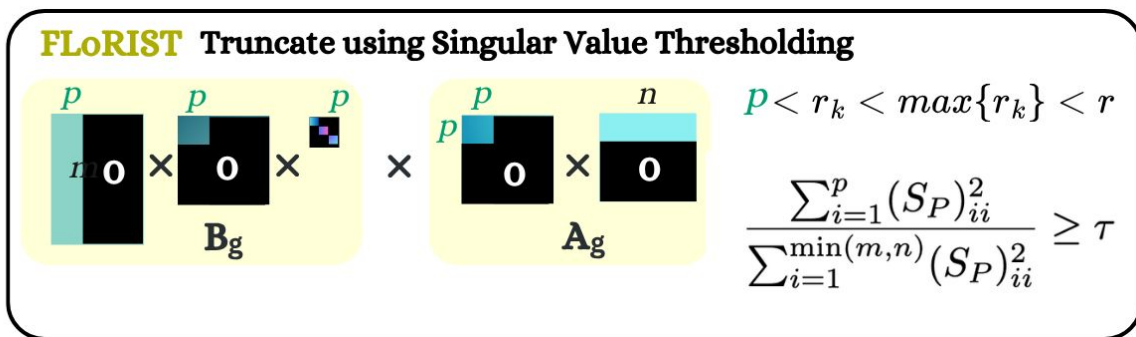
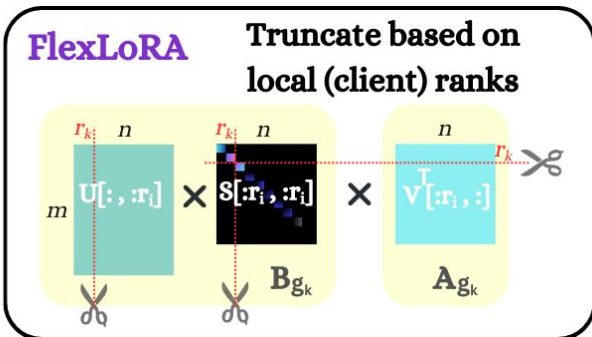
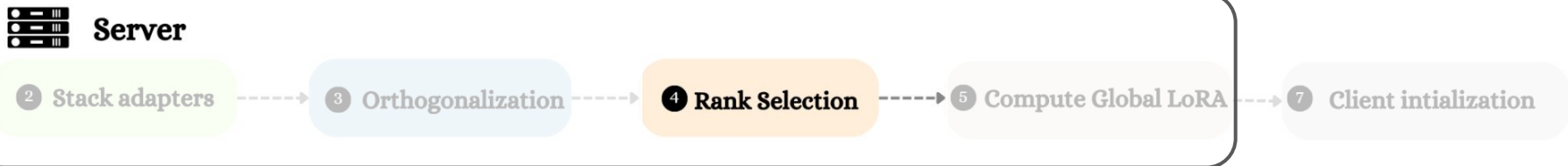


Challenge 2: SVD based orthogonalization is computationally expensive

Bai et al., "Federated Fine-Tuning of Large Language Models under Heterogeneous Tasks and Client Resources," NeurIPS 2024.

Ramesh, H., Dass, J.: FLoRIST: Singular value thresholding for efficient and accurate federated fine-tuning of large language models. MLSys 2026

Related Work: Stacking

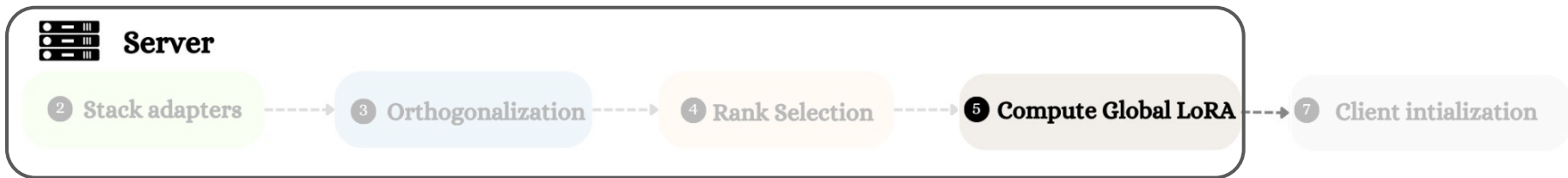


Challenge 3: Heuristic-based rank selection methods

Bai et al., "Federated Fine-Tuning of Large Language Models under Heterogeneous Tasks and Client Resources," NeurIPS 2024.

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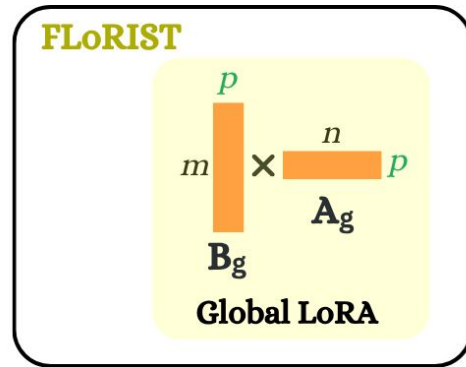
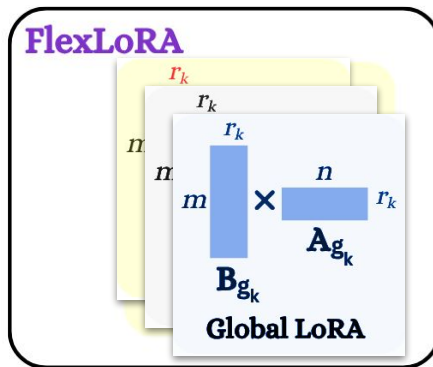
Related Work: Stacking



BROADCAST:

FlexLoRA: Separate adapter to every client

FLoRIST: Single shared adapter to all clients



Related Work: Stacking



FlexLoRA

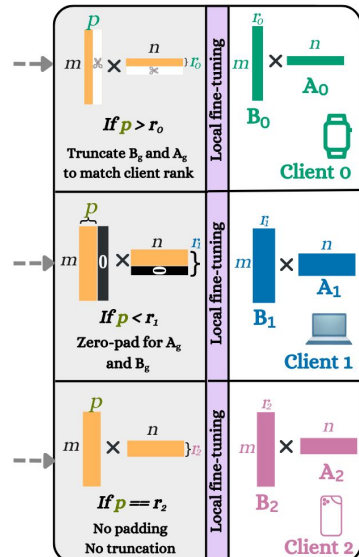
$$\begin{matrix} r_k \\ m \end{matrix} \times \begin{matrix} n \\ r_k \end{matrix}$$

$$\begin{matrix} B_{g_k} \\ A_{g_k} \end{matrix}$$

Clients initialize with global LoRA without any additional truncation/padding

Challenge 4: Zero-pad initialization limits exploration

FLoRIST



Bai et al., "Federated Fine-Tuning of Large Language Models under Heterogeneous Tasks and Client Resources," NeurIPS 2024.

Ramesh, H., Dass, J.: FLoRIST: Singular value thresholding for efficient and accurate federated fine-tuning of large language models. MLSys 2026

Summarizing Challenges



Challenge 1: Stacking based methods haven't been explored in Vision



Challenge 2: SVD based orthogonalization is computationally expensive



Challenge 3: Heuristic-based rank selection methods



Challenge 4: Zero-pad initialization limits exploration

Zhang et al., "Towards Building the Federated GPT: Federated Instruction Tuning," ICASSP 2024.

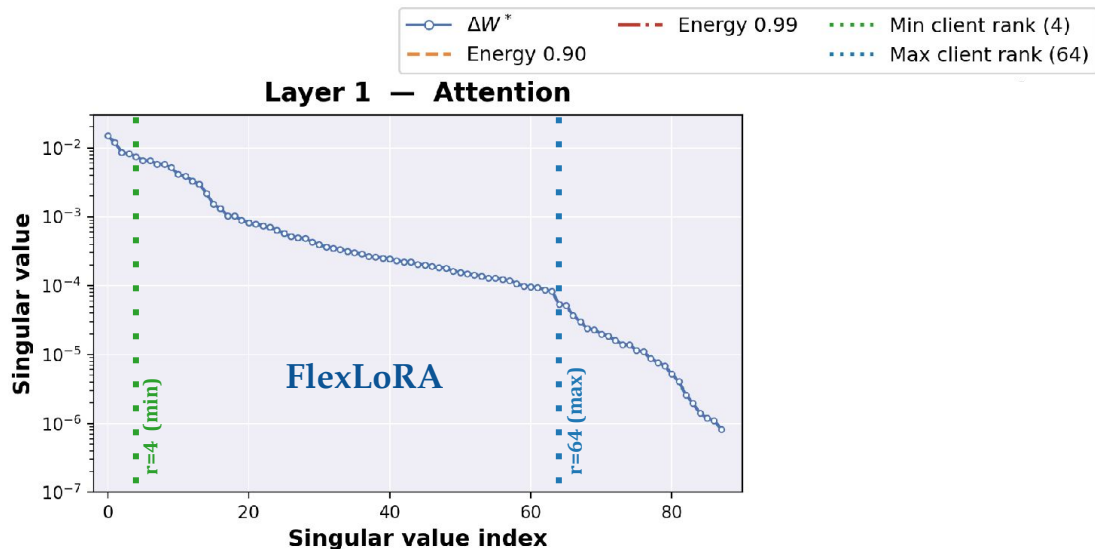
Bian et al., "LoRA-FAIR: Federated LoRA Fine-Tuning with Aggregation and Initialization Refinement," ICCV 2025.

Bai et al., "Federated Fine-Tuning of Large Language Models under Heterogeneous Tasks and Client Resources," NeurIPS 2024.

Ramesh, H., Dass, J.: FLoRIST: Singular value thresholding for efficient and accurate federated fine-tuning of large language models. MLSys 2026

Understanding Global Update In Vision Fine-Tuning

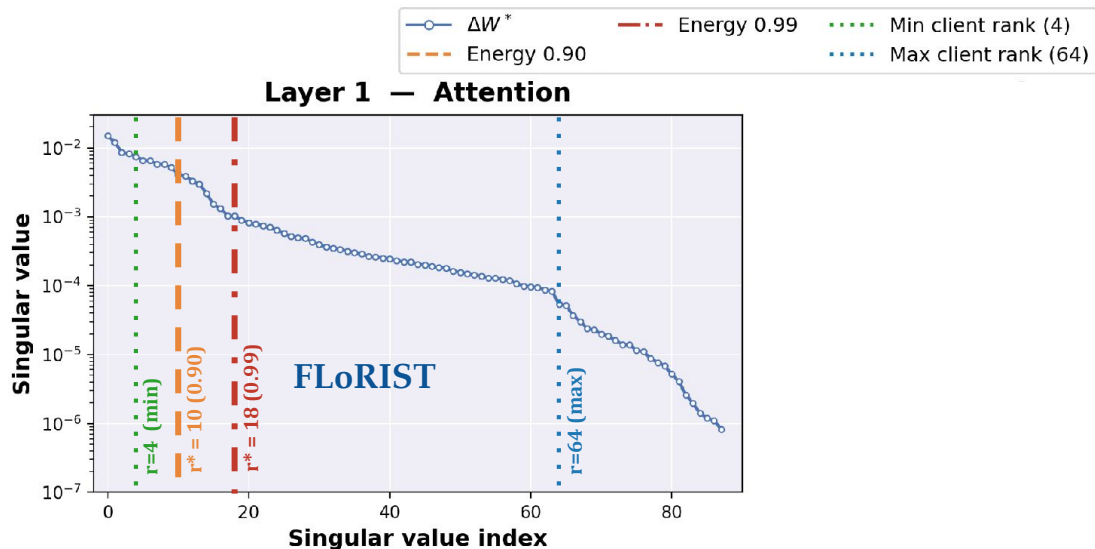
Singular value spectrum of ΔW , ViT-L/16 on NICO++ (Heterogeneous ranks)



Our Motivating Empirical Study

Understanding Global Update In Vision Fine-Tuning

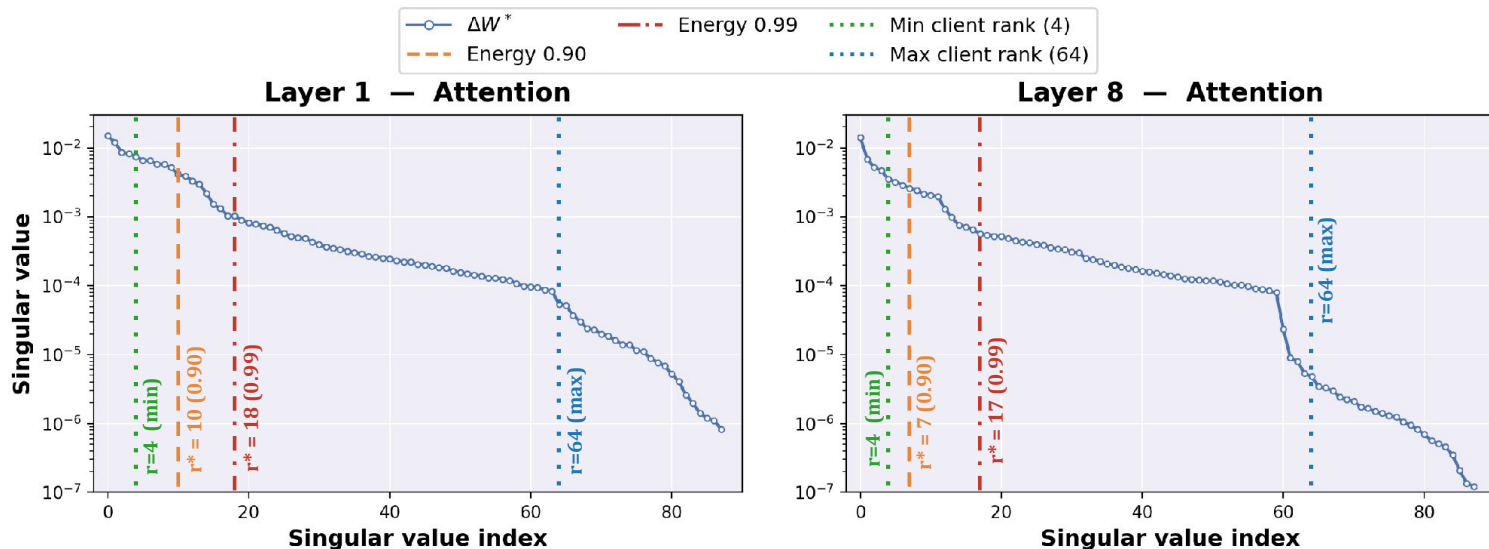
Singular value spectrum of ΔW , ViT-L/16 on NICO++ (Heterogeneous ranks)



Our Motivating Empirical Study

Understanding Global Update In Vision Fine-Tuning

Singular value spectrum of ΔW , ViT-L/16 on NICO++ (Heterogeneous ranks)



Our Motivating Observation: Optimal rank varies across Layers

Heuristic-based Rank Selection Thresholds are unreliable!

No single τ works: $\tau = 0.90$ cuts into signal; $\tau = 0.99$ keeps noise

FLoRIST's own optimal τ varies between 0.80–0.99 across models/datasets



Research Question

For federated LoRA fine-tuning in vision models

How to design stack-based aggregation which automatically identifies optimal layer-wise LoRA ranks for global model update without manually tuning threshold ?

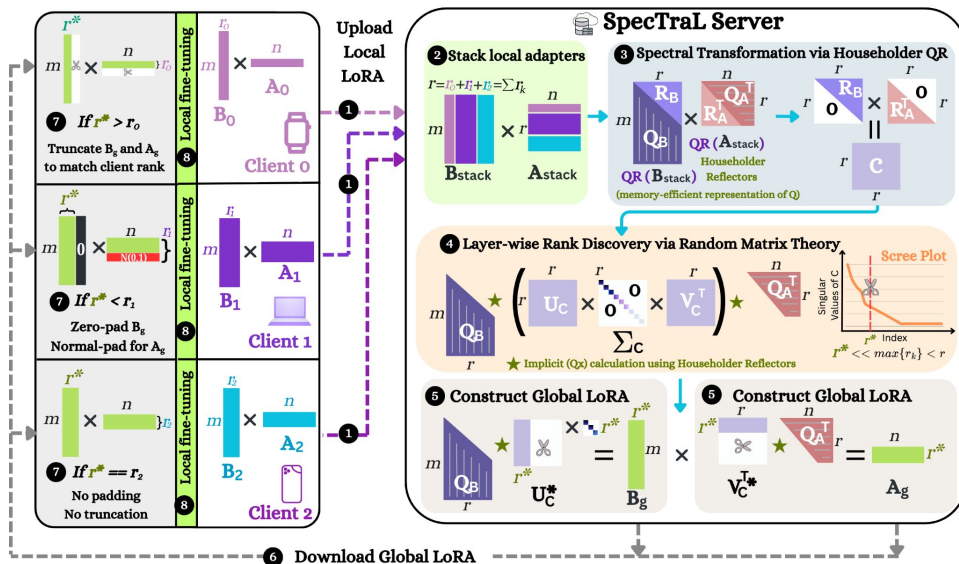
Proposed Method: SpecTraL

For federated LoRA fine-tuning in vision models

How to design stack-based aggregation which automatically identifies optimal layer-wise LoRA ranks for global model update without manually tuning threshold?

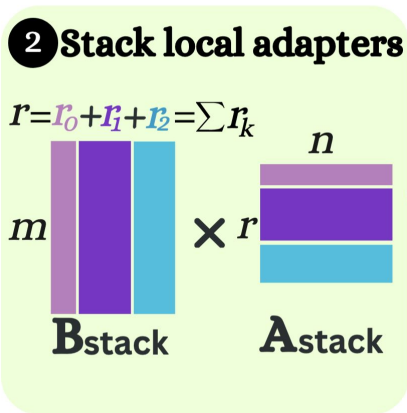
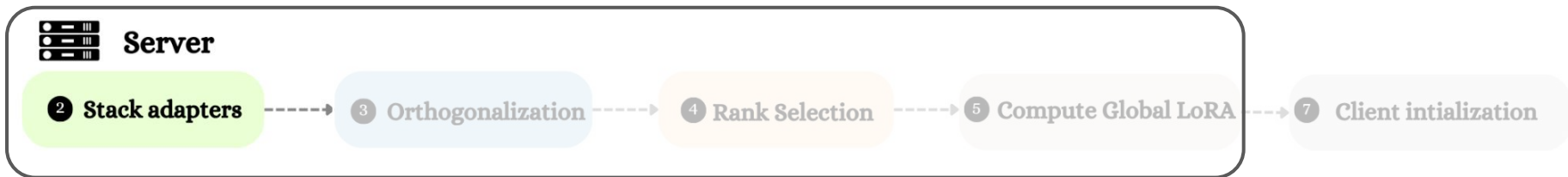
- ! Challenge 1: Stacking based methods haven't been explored in Vision
- ! Challenge 2: SVD based orthogonalization is computationally expensive
- ! Challenge 3: Heuristic-based rank selection methods
- ! Challenge 4: Zero-pad initialization limits exploration

Spectral Transformation for Layer-wise Global Rank Discovery for ViT

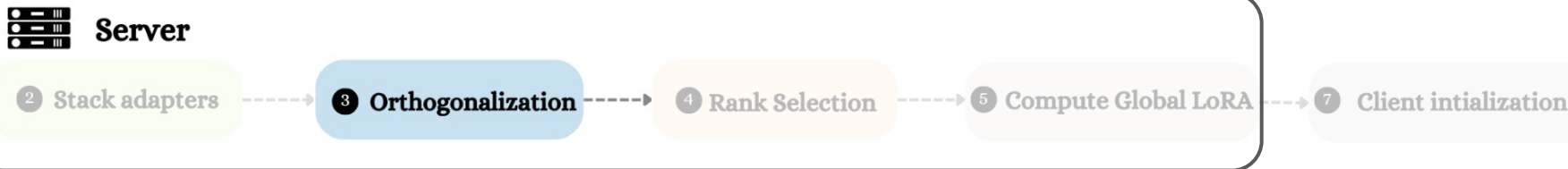


-  **Householder-QR aggregation**
Exact singular values of ΔW^* without reconstruction
Faster than FLoRIST/FlexLoRA
-  **Principled denoising (ScreeNOT)**
Separates signal from non-IID noise
No threshold tuning per-layer
-  **Gaussian initialization for padding**
Promotes exploratory directions for local updates
Faster convergence

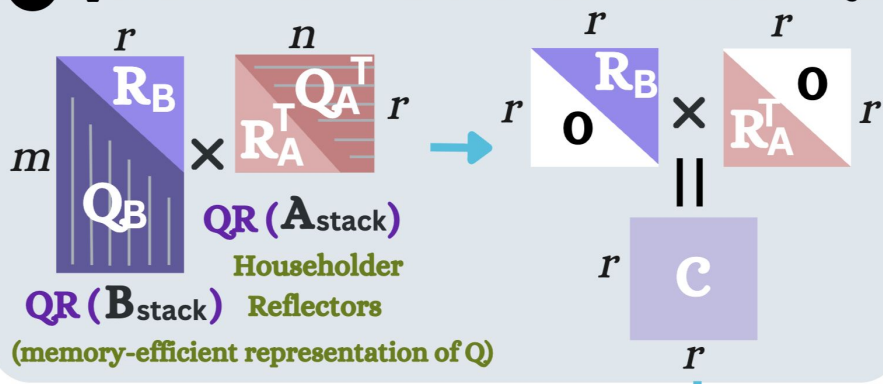
Proposed Method: SpecTraL



Proposed Method: SpecTraL



3 Spectral Transformation via Householder QR



$$C = R_B R_A^T \in \mathbb{R}^{r \times r}$$

$$\Delta W^* = Q_B R_B (Q_A R_A)^T = Q_B \underbrace{(R_B R_A^T)}_C Q_A^T$$

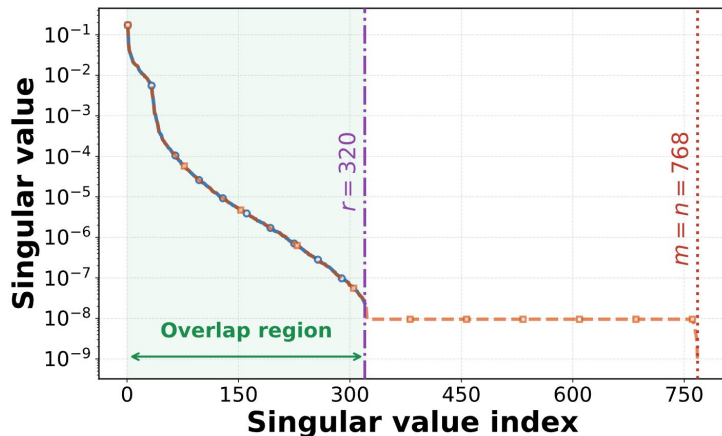
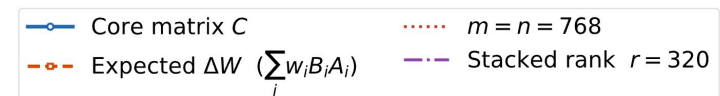
$$\text{SVD}(C) = U_C \Sigma_C V_C^T$$

$$\Delta W^* = (Q_B U_C) \Sigma_C (V_C^T R_A^T)$$

Proposed Method: SpecTraL



Server



$$\Delta W^* = (Q_B U_C) \Sigma_C (V_C^T R_A^T) \quad SVD(\Delta W) = U S V^T$$

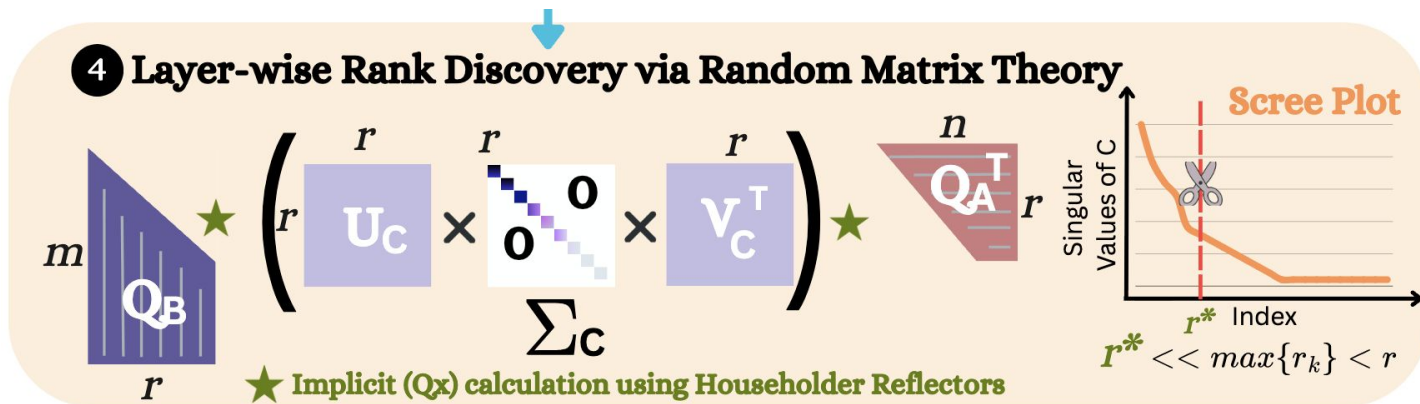


*Singular values (C) == Singular values (ΔW)
without global update reconstruction*

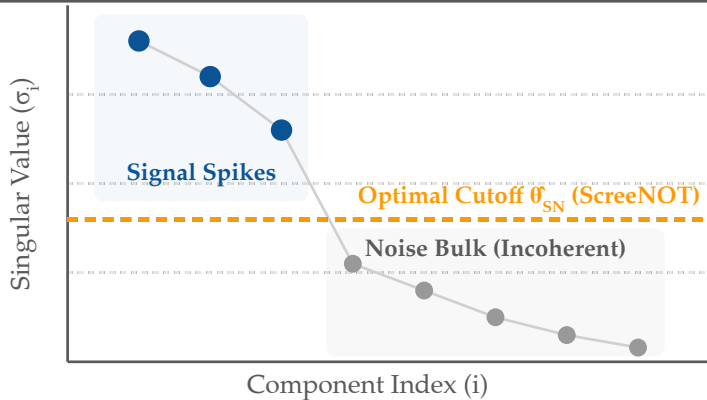
Householder QR orthogonalizes in compact latent space,
recovering the exact spectrum of ΔW^* without a full SVD

~7× faster than FLoRIST's SVD

Proposed Method: SpecTraL



Proposed Method: SpecTraL



ScreenNOT Selection Rule

$$\hat{\sigma}_i = \sigma_i \text{ if } \sigma_i > \hat{\theta}_{SN}, \text{ else } 0$$

$$r^* = |\{i : \sigma_i > \hat{\theta}_{SN}\}|$$

How is θ_{SN}^* identified?

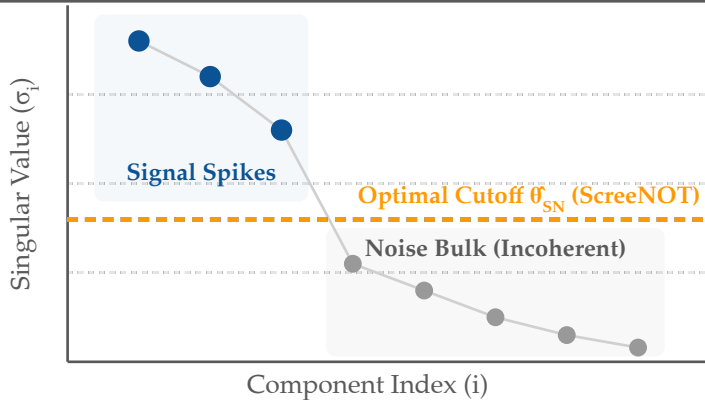
Hard-threshold reconstruction of the signal

$$\Delta \hat{W}(\theta) = \sum_{i: \sigma_i > \theta} \sigma_i \mathbf{u}_i \mathbf{v}_i^T$$

The threshold minimizes the signal-reconstruction error (MSE):

$$\theta_{SN}^* = \operatorname{argmin}_{\theta} \|\Delta \hat{W}(\theta) - \Delta W_{\text{signal}}\|_F^2$$

Proposed Method: SpecTraL



ScreeNOT Selection Rule

$$\hat{\sigma}_i = \sigma_i \text{ if } \sigma_i > \hat{\theta}_{SN}, \text{ else } 0$$

$$r^* = |\{i : \sigma_i > \hat{\theta}_{SN}\}|$$

Handles correlated noise

Non-IID clients produce **correlated noise**;

Spiked-covariance model

Signal spikes sit above Marchenko–Pastur noise bulk, Random Matrix Theory identifies this pattern.

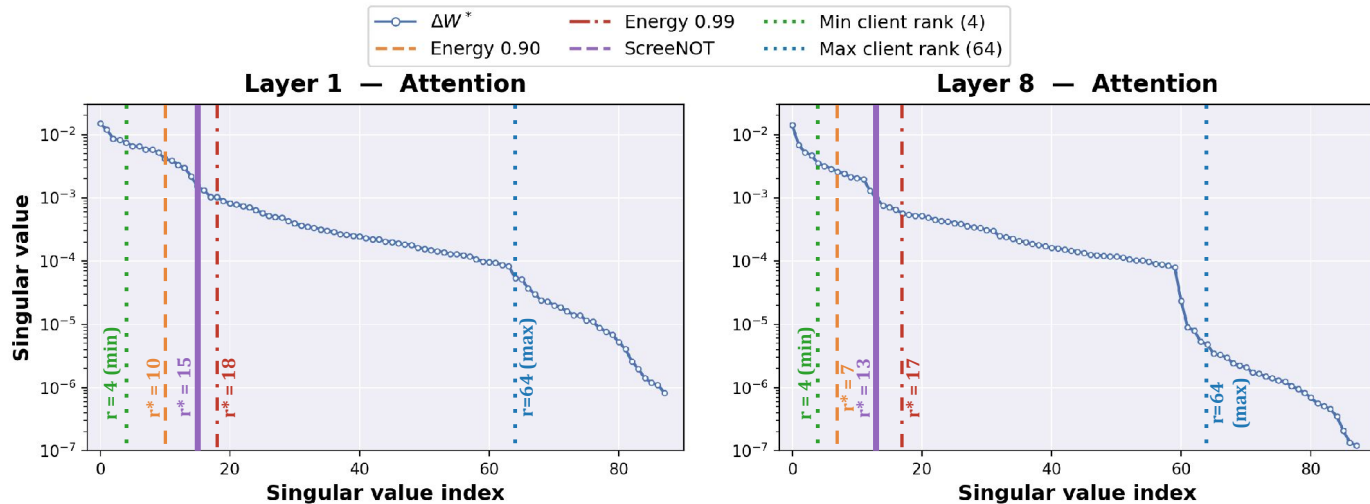
Optimal Cutoff Identification

ScreeNOT returns the **MSE-optimal hard threshold** that separates signal from noise, with no manual tuning.

Proposed Method: SpecTraL

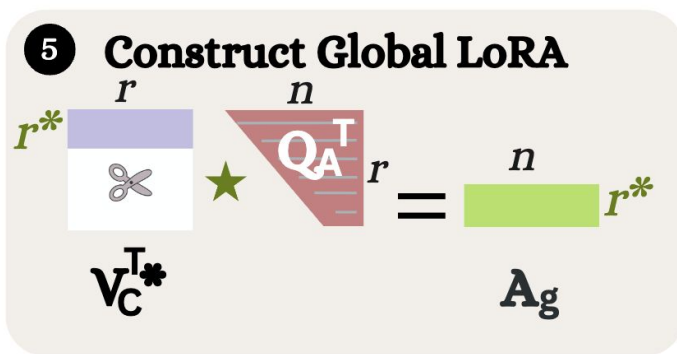
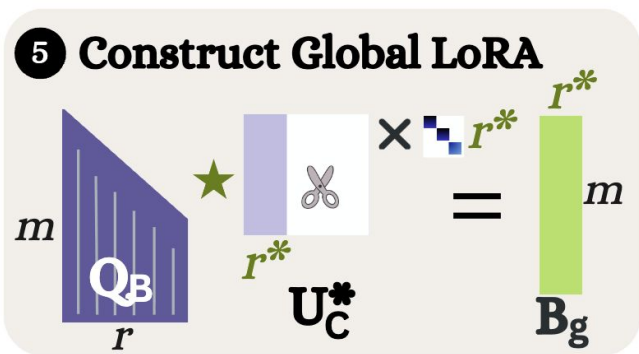


Server



ScreeNOT identifies the optimal threshold for elbow separating signal from noise without any manual tuning

Proposed Method: SpecTraL



Global adapters are built by applying stored Householder reflectors to the top- r^* columns, never forming the dense update.

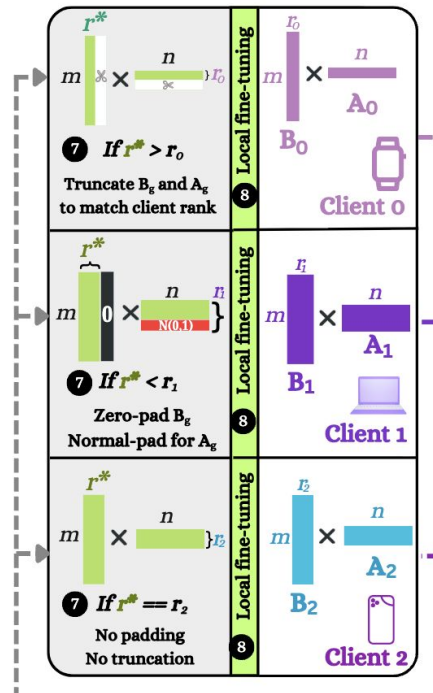
Proposed Method: SpecTraL



2 Stack adapters -----> 3 Orthogonalization -----> 4 Rank Selection -----> 5 Compute Global LoRA -----> 7 Client initialization

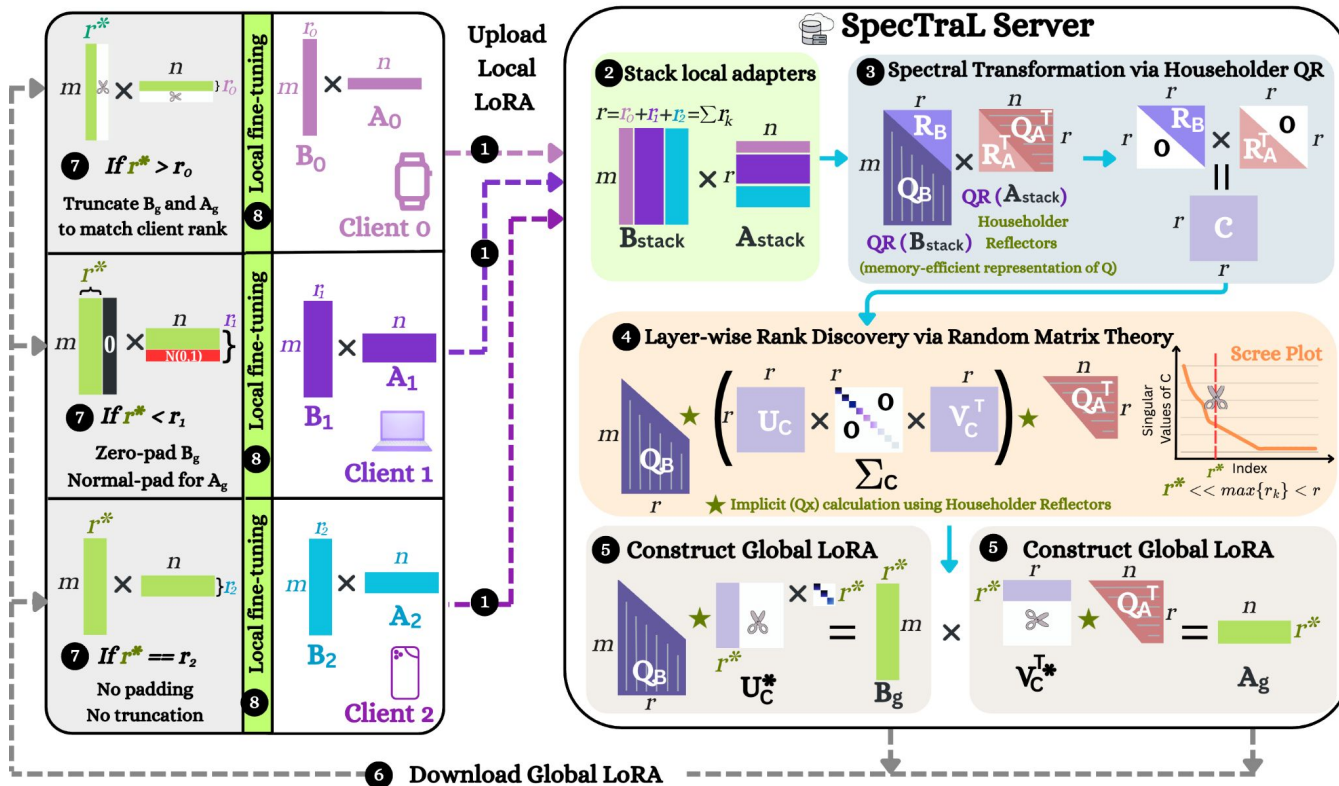


Gaussian residual init preserves the denoised signal exactly while giving new dimensions gradient from step one, outperforming zero-padding.



Proposed Method: SpecTraL

Spectral Transformation for Layer-wise Global Rank Discovery



Experimental Setup

Model Architectures

ViT-B/16 & ViT-L/16 pre-trained on ImageNet-21k.
Adapting attention projection and MLP layers.

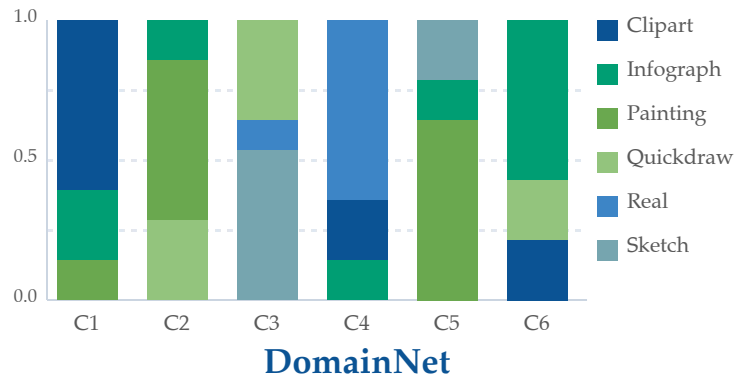
Real-World Benchmarks

- **DomainNet (Multi-Domain Image Classification):** clipart, infograph, painting, quickdraw, real, sketch
- **NICO++ (Multi-Domain Image Classification):** autumn, dim, grass, outdoor, rock, water

Federated Environment

- **Scale:** 100 clients total, 10 sampled per round
- **Partitioning:** Feature & label non-IID split
- **Skew Parameter:** Dirichlet distribution ($\alpha = 0.5$)

DIRICHLET LABEL DISTRIBUTION SKEW ($\alpha = 0.5$)



Evaluation Summary

Rigorous testing over 12 distinct domains across DomainNet and NICO++ confirms SpecTraL's robustness in heavily skewed federated environments.

Results: Accuracy across Models & Datasets

Table 1: Per-domain accuracy (%) under feature and label non-IID with client rank $r = 32$ (100 clients, 75 rounds). Best in **bold**, second-best underlined. SpecTraL uses ScreeNOT + Gaussian initialization.

Model	Dataset	Method	Per-Domain Accuracy (%)						Avg.
ViT-L/16	DomainNet		Clipart	Infograph	Painting	Quickdraw	Real	Sketch	
		FFA-LoRA	77.75	45.18	79.82	55.87	88.86	71.45	69.82
		FedIT	84.23	58.06	84.30	76.35	91.11	81.50	79.26
		FLoRA	77.96	44.74	80.11	59.95	89.27	71.68	70.62
		FlexLoRA	<u>84.88</u>	<u>58.32</u>	84.25	<u>76.54</u>	<u>91.31</u>	<u>81.58</u>	<u>79.48</u>
		LoRA-FAIR	84.41	57.88	84.22	76.16	91.11	81.37	79.19
		FLoRIST	84.60	57.09	84.25	75.51	91.11	80.92	78.91
		SpecTraL	85.07	58.97	84.36	76.64	91.32	81.74	79.68
	NICO++		Autumn	Dim	Grass	Outdoor	Rock	Water	
		FFA-LoRA	91.20	87.79	92.62	90.53	89.60	89.22	90.16
		FedIT	93.51	90.75	94.56	91.82	92.27	91.69	92.43
		FLoRA	91.42	87.87	92.69	90.44	89.24	88.88	90.09
		FlexLoRA	94.00	92.51	95.17	92.48	92.36	92.58	93.18
		LoRA-FAIR	90.87	85.94	93.26	88.94	88.79	88.96	89.46
		FLoRIST	93.73	92.03	94.44	92.24	92.45	92.11	92.83
		SpecTraL	93.73	<u>92.39</u>	<u>95.13</u>	92.86	92.95	93.02	93.35



SpecTraL consistently achieves the highest average accuracy across all four model-dataset combinations; while FlexLoRA is consistently the 2nd strongest

Results: Server Compute & Communication

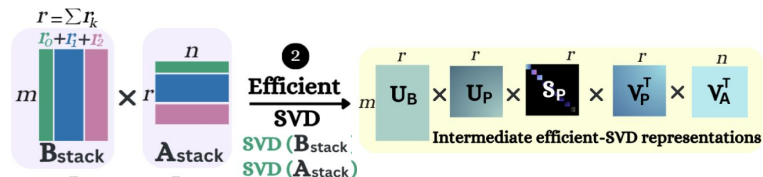
Table 3: Server-side computation and per-client download communication per round on ViT-L/16 (NICO++) under heterogeneous client ranks, with $K = 10$ clients sampled per round, per-client rank $r_k = 32$ (stacked rank $r = 320$), measured at round 1.

Metric	FedIT	FFA-LoRA	FLoRA	FlexLoRA	LoRA-FAIR	FLoRIST	SpecTraL
Server Comp. (GFLOPs) ↓	0.14	0.07	0	6212.54	38848.12	824.38	200.04
Download Comm. (MB) ↓	54.00	27.00	540.00	54.00	54.00	8.67	7.42



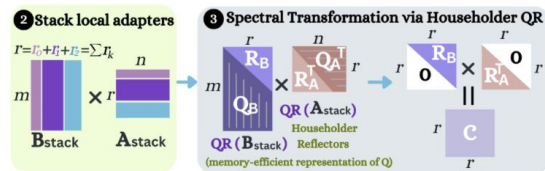
SpecTraL is 4.1× / 31× / 194× faster than FLoRIST / FlexLoRA / LoRA-FAIR

SpecTraL downloads just 7.42 MB/client, the smallest of all methods



FLoRIST

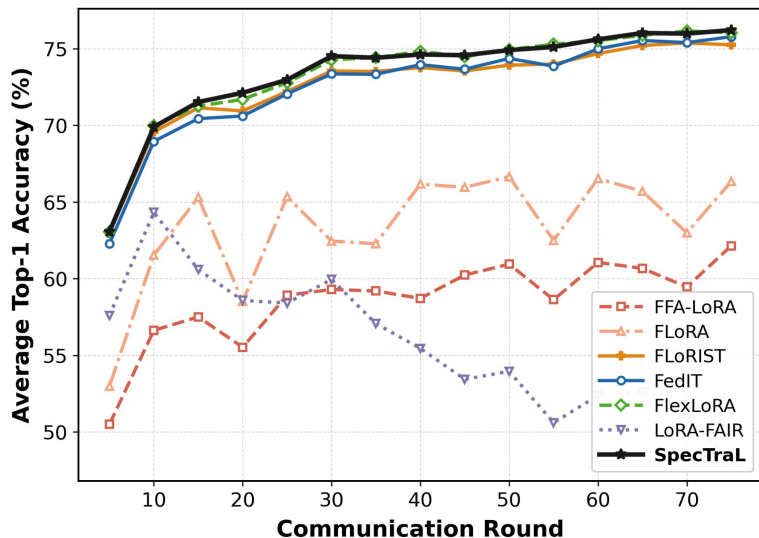
SpecTraL speedup over FLoRIST comes from Householder-QR



SpecTraL

Results: Convergence

ViT-B/16-DomainNet-homogeneous

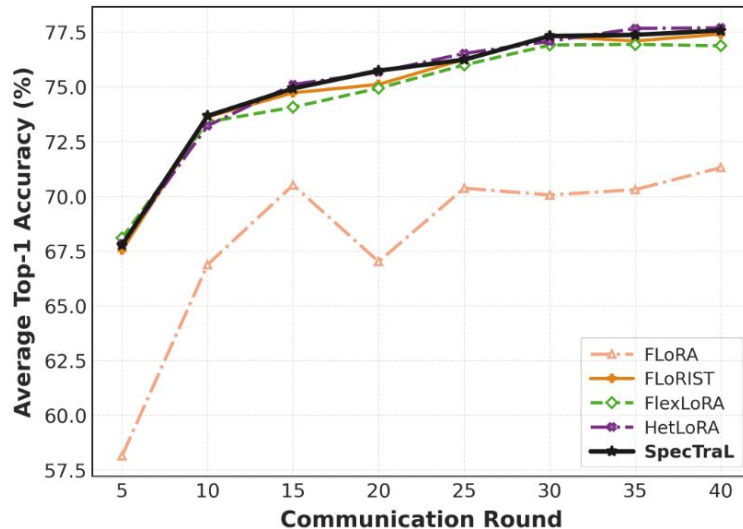


All clients rank $r = 32$

SpecTraL achieves **higher accuracy throughout training**;
Converges at least as fast as all baselines,
operating entirely in compact $r \times r$ space



ViT-L/16-DomainNet-heterogeneous



Client ranks: $40 \times r=4$, 20×8 , 20×16 , 10×32 , 10×64

SpecTraL matches/exceeds baseline convergence ($\sim 77-78\%$)
even with **client ranks varying by a factor of 16**

Ablation: Initialization Strategies for SpecTraL

Table 2: Ablation of various initialization strategies with ScreenNOT thresholding on DomainNet (ViT-B/16, $r=32$, 75 rounds). Best and second-best results are **bolded** and underlined.

Initialization	Clipart	Infograph	Painting	Quickdraw	Real	Sketch	Avg.
Zero-padding	82.11	51.70	79.92	74.10	90.18	77.73	75.96
Gaussian	<u>82.45</u>	<u>51.67</u>	79.95	74.30	90.60	78.27	76.21
Trained-A	81.61	51.17	79.48	73.76	90.48	77.81	75.72
Pretrained-SVD	82.68	52.44	79.39	<u>74.26</u>	90.46	<u>77.92</u>	<u>76.19</u>
Orthogonal compl.	82.29	51.94	<u>79.74</u>	74.14	<u>90.72</u>	77.61	76.07



All five SpecTraL initialization strategies beat FLoRIST (75.25%) which uses Zero-padding
SpecTraL (with Gaussian init.) wins (76.21%) with best exploration
Zero-padding is the weakest at 75.96%.

Conclusion

1. **SpecTraL brings stacking-based aggregation to Vision Transformers**, exact and heterogeneity-native.
2. **SpecTraL replaces expensive SVD with Householder QR**, recovering the exact spectrum at a fraction of the cost.
3. **SpecTraL replaces heuristic thresholds** with ScreeNOT's MSE-optimal, layer-adaptive rank.
4. **SpecTraL initializes residuals with Gaussian padding**, outperforming zero-padding.
5. **Result: highest accuracy at lower server cost**, $4.1\times$ cheaper than FLoRIST.



Acknowledgments



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Appendix

Additional Results: Accuracy across Models & Datasets

Table A1: Per-domain accuracy (%) under feature and label non-IID with client rank $r = 32$ (100 clients, 75 rounds). Best in **bold**, second-best underlined. SpecTraL uses ScreeNOT + Gaussian initialization.

	Model	Dataset	Method	Per-Domain Accuracy (%)						Avg.
				Clipart	Infograph	Painting	Quickdraw	Real	Sketch	
ViT-B/16	DomainNet	FFA-LoRA		70.18	39.41	74.50	40.28	87.85	60.59	62.13
		FedIT		82.01	51.91	<u>79.88</u>	72.64	90.86	77.34	75.77
		FLoRA		73.02	40.50	75.58	53.45	<u>88.69</u>	66.73	66.33
		FlexLoRA		81.98	51.64	79.60	<u>74.10</u>	90.63	<u>78.27</u>	<u>76.04</u>
		LoRA-FAIR		53.02	21.94	47.96	55.61	68.62	42.09	48.21
		FLoRIST		81.80	50.20	79.10	72.69	90.35	77.34	75.25
		SpecTraL		82.45	51.67	79.95	74.30	90.60	78.27	76.21
	NICO++			Autumn	Dim	Grass	Outdoor	Rock	Water	
		FFA-LoRA		89.33	83.46	91.26	88.04	87.52	87.34	87.83
		FedIT		90.65	88.39	93.16	89.99	<u>89.78</u>	<u>90.60</u>	90.43
		FLoRA		89.49	83.94	91.39	88.31	88.34	87.50	88.16
		FlexLoRA		<u>91.03</u>	88.91	<u>93.67</u>	<u>90.71</u>	89.56	90.60	<u>90.75</u>
		LoRA-FAIR		74.71	69.61	79.56	77.53	74.64	76.07	75.35
		FLoRIST		91.14	<u>88.83</u>	93.77	90.44	89.83	89.76	90.63
		SpecTraL		91.58	88.19	93.65	90.86	89.92	90.96	90.86

Table A2: Per-domain accuracy (%) under feature and label non-IID with client rank $r = 32$ (100 clients, 75 rounds). Best in **bold**, second-best underlined. SpecTraL uses ScreeNOT + Gaussian initialization.

	Model	Dataset	Method	Per-Domain Accuracy (%)						Avg.
				Clipart	Infograph	Painting	Quickdraw	Real	Sketch	
ViT-L/16	DomainNet	FFA-LoRA		77.75	45.18	79.82	55.87	88.86	71.45	69.82
		FedIT		84.23	58.06	<u>84.30</u>	76.35	91.11	81.50	79.26
		FLoRA		77.96	44.74	80.11	59.95	89.27	71.68	70.62
		FlexLoRA		<u>84.88</u>	<u>58.32</u>	84.25	<u>76.54</u>	<u>91.31</u>	<u>81.58</u>	<u>79.48</u>
		LoRA-FAIR		84.41	57.88	84.22	76.16	91.11	81.37	79.19
		FLoRIST		84.60	57.09	84.25	75.51	91.11	80.92	78.91
		SpecTraL		85.07	58.97	84.36	76.64	91.32	81.74	79.68
	NICO++			Autumn	Dim	Grass	Outdoor	Rock	Water	
		FFA-LoRA		91.20	87.79	92.62	90.53	89.60	89.22	90.16
		FedIT		93.51	90.75	94.56	91.82	92.27	91.69	92.43
		FLoRA		91.42	87.87	92.69	90.44	89.24	88.88	90.09
		FlexLoRA		94.00	92.51	95.17	92.48	92.36	92.58	93.18
		LoRA-FAIR		90.87	85.94	93.26	88.94	88.79	88.96	89.46
		FLoRIST		93.73	92.03	94.44	92.24	92.45	92.11	92.83
		SpecTraL		93.73	<u>92.39</u>	<u>95.13</u>	92.86	92.95	93.02	93.35

SpecTraL consistently achieves the highest average accuracy across all four model-dataset combinations; while FlexLoRA is consistently the 2nd strongest

Results: Impact of non-iid heterogeneity

NICO++, homogeneous ranks, 100 clients, 75 rounds, round-75 accuracy over 3 seeds.

Table : Robustness across data heterogeneity on NICO++ (homogeneous client ranks, $r = 32$, 100 clients, sample fraction 0.1, 75 rounds). Average accuracy (%) at standard ($\alpha = 0.5$) and extreme ($\alpha = 0.1$) label heterogeneity. Best in **bold**, second-best underlined. SpecTraL uses ScreeNOT + Gaussian initialization.

Method	ViT-B/16		ViT-L/16	
	$\alpha = 0.5$	$\alpha = 0.1$	$\alpha = 0.5$	$\alpha = 0.1$
FFA-LoRA	87.83	87.41 ± 0.11	90.16	88.86 ± 0.36
FedIT	90.43	<u>90.10</u> ± 0.11	92.43	91.82 ± 0.18
FLoRA	88.16	87.58 ± 0.09	90.09	89.07 ± 0.63
FlexLoRA	<u>90.75</u>	90.09 ± 0.25	<u>93.18</u>	<u>92.18</u> ± 0.15
LoRA-FAIR	75.35	53.91 ± 3.42	89.46	84.96 ± 1.14
FLoRIST	90.63	89.60 ± 0.12	92.83	91.88 ± 0.22
SpecTraL	90.86	90.12 ± 0.07	93.35	92.24 ± 0.08



SpecTraL stays the best method as heterogeneity intensifies ($\alpha: 0.5 \rightarrow 0.1$), on both ViT-B/16 and ViT-L/16, with very low variance (± 0.07 – 0.08).

Additional Results: Generality to Self-Supervised Backbones

NICO++, $\alpha = 0.1$, homogeneous ranks, 100 clients, 75 rounds, round-75 accuracy over 3 seeds.

Table A4: Generality to *self-supervised* backbones: DINOv2 (ViT-B/14 and ViT-L/14 encoders) on NICO++ under extreme heterogeneity ($\alpha = 0.1$, homogeneous ranks, $r = 32$, 100 clients, fraction 0.1, 75 rounds). Round-75 accuracy (%), mean $\pm$ std over seeds {42,43,44}. Best in **bold**, second-best underlined.

Method	DINOv2-B	DINOv2-L
FFA-LoRA	92.14 $\pm$ 0.20	93.83 $\pm$ 0.13
FedIT	<u>93.67</u> $\pm$ 0.20	94.78 $\pm$ 0.13
FLoRA	92.03 $\pm$ 0.52	93.86 $\pm$ 0.09
FlexLoRA	93.69 $\pm$ 0.10	95.08 $\pm$ 0.20
LoRA-FAIR	4.52 $\pm$ 0.28	2.80 $\pm$ 0.88
FLoRIST	93.39 $\pm$ 0.16	94.76 $\pm$ 0.23
SpecTraL	93.46 $\pm$ 0.29	<u>95.02</u> $\pm$ 0.09

SpecTraL transfers beyond supervised ViTs to self-supervised DINOv2, staying within noise of the best method on both Base (93.46) and Large (95.02).

SpecTraL still matches FlexLoRA (95.02 vs 95.08 on Large) at a fraction of its server cost.

Additional Results: Generality to Language Models

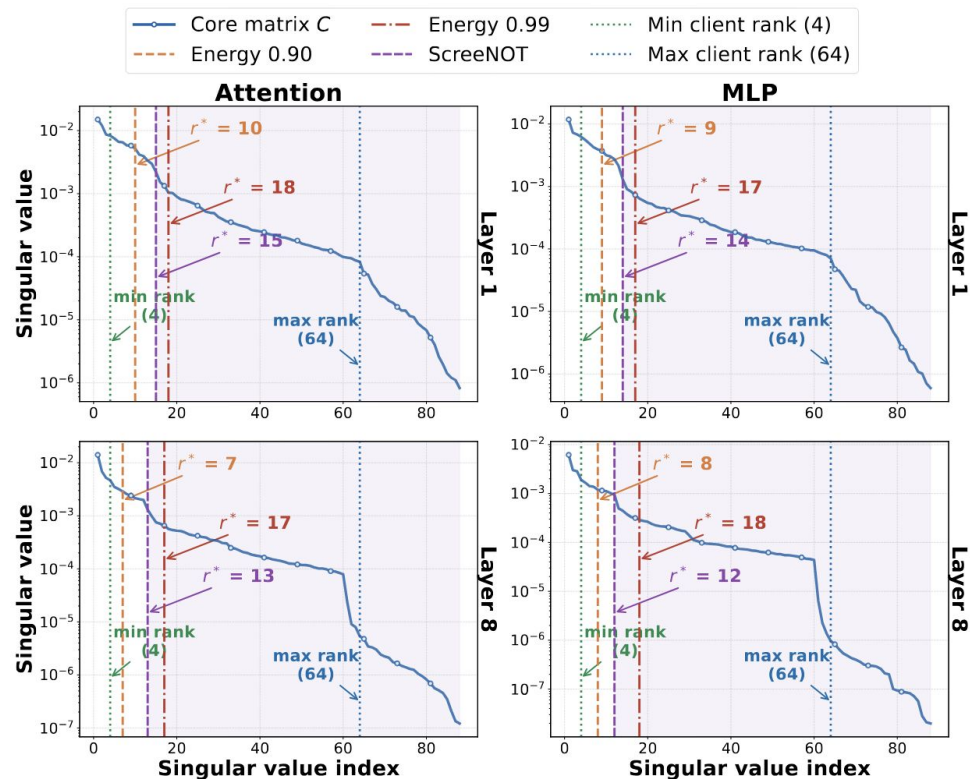
RoBERTa, homogeneous ranks $r=32$, 50 clients, fraction 0.2, final-round accuracy, mean over 4 seeds.

Table A5: Generality to language models: RoBERTa on GLUE **SST-2** at low ($\alpha = 0.2$) and mild ($\alpha = 1.0$) data heterogeneity. Homogeneous ranks, $r = 32$, 50 clients, fraction 0.2, SGD. Final-round (r150) accuracy (%), mean $\pm$ std over 4 seeds. Best in **bold**, second-best underlined.

Method	$\alpha = 0.2$	$\alpha = 1.0$
FFA-LoRA	<u>93.96</u> ± 0.80	95.68 ± 0.24
FedIT	82.71 ± 19.42	95.67 ± 0.10
FLoRA	93.88 ± 0.80	95.72 ± 0.19
FlexLoRA	93.09 ± 1.37	<u>95.90</u> ± 0.45
LoRA-FAIR	93.54 ± 0.71	95.64 ± 0.16
FLoRIST	93.27 ± 0.35	95.60 ± 0.44
SpecTraL	94.15 ± 1.28	96.04 ± 0.42

SpecTraL transfers from vision to language,
staying in the top tier on SST-2 across all heterogeneity levels.

Fixed thresholds are unreliable



Singular value spectrum of core matrix C, ViT-L/16 on NICO++(Heter)

Our Motivating Observation

Optimal rank varies across layers & weight types (r^* ranges 7 $\rightarrow$ 18)

No single τ works: $\tau = 0.90$ cuts into signal; $\tau = 0.99$ keeps noise

FLoRIST's own optimal τ varies between 0.80–0.99 across models/datasets